Himanshu Kukreja

Senior Software Engineer & Engineering Lead · Senseloaf AI

kukreja.him@gmail.com
himanshukukreja.in
github.com/himanshkukreja
linkedin.com/in/himanshukukreja
x.com/hi_kukreja · Bengaluru, India

Engineering lead who owns product **end to end**: product research, UI design, frontend, backend and infra. Founding engineer twice (AutoFlow, NativeBridge). Builds **agentic AI products**, **real-time device streaming over WebRTC** and **developer tools**; ships products of his own on the side (**Chukta**, **Markdrop**), open-sources the tools and teaches system design to backend engineers.

EXPERIENCE

Senior Software Engineer & Engineering Lead · Senseloaf AI
Apr 2026 – Present · Bengaluru

- Lead engineering of an **agentic hiring platform** used by enterprises and startups, with AI agents for **sourcing, resume-to-job matching, profile verification, prescreening and AI interviews**.
- Building **SIA (Senseloaf Intelligent Agent)**, a chat interface where a recruiter runs hiring end to end from one conversation, and the **MCP server** behind it.
- Own features end to end: product research, UI design, frontend, backend and **infra scaling**.

Software Engineer (Freelance) · pCloudy
Mar 2026 – Jun 2026 · Remote

- Researched **device streaming** for a cloud of **5,000+** devices, and built mobile app testing solutions.

Founding Engineer · NativeBridge
Jan 2025 – Mar 2026 · Bengaluru

- Built cloud **Android and iOS devices streamed to the browser over WebRTC** at **<100 ms**, with remote touch, rotation, clipboard sync and file transfer.
- Created **iOS Bridge** (open source, **75★**, 10 forks): a **FastAPI** server on macOS that drives iOS simulators through **simctl and idb** and streams them to Electron desktop apps (Windows, Linux, macOS), a browser client and a pip-installable CLI. Sessions survive server restarts; app install, MP4 recording and location simulation over REST and WebSocket APIs.
- Replaced screenshot-over-WebSocket streaming (**2–5 FPS**, ~2 s lag) with a **WebRTC video track** fed from a one-frame buffer that always keeps the newest frame: **60–90 FPS** at **50–100 ms**.
- Built **BridgeLink** (open source, on PyPI and Homebrew): a Python CLI that exposes USB or Wi-Fi Android devices to the NativeBridge platform through **API-key-authenticated bore tunnels**, so a remote developer runs **adb** as if the phone were plugged in. Background daemons health-check every device **each second** and re-activate it when it reconnects.

Founding Engineer #1 · AutoFlow
Jan 2024 – Dec 2024 · Bengaluru

- Employee #1: built the **whole backend from an empty repo** in **FastAPI** on PostgreSQL, plus a **Bun WebSocket gateway**.
- Shipped **AI-driven autonomous UI testing**: LLM agents that write and run UI tests on a fleet of **Playwright runners**, with runner workers across **AWS and GCP**.
- Launched on Product Hunt as **Product of the Day**.

Software Development Engineer 1 · Ikarus 3D
Jul 2023 – Oct 2023 · Chandigarh

- Led design and development of **3 business-critical applications**: carts and orders, the authentication system and the FDM slicer.
- Migrated the online FDM slicer from **AWS ECS and EFS to GCP Cloud Run**, solving runtime vertical scaling and cutting its cost by **90%**.
- Set up the company's **first CI/CD pipelines** for business applications: faster deploys, fewer build failures and integration issues.
- Extended authentication with **device-ID tracking**, new sign-in notifications, password recovery and email change.
- Researched **3D generative AI** with Dr. Sahil Sharma, turning text and 2D images into 3D models with NeRF, BLIP and DreamFusion: **48%** accuracy within 30 days.
- Automated the company's **HRMS** with biometric hardware integration over APIs and webhooks, saving **132 hours** a year of HR work; mentored **3 interns**.

Software Developer Associate · Ikarus 3D
Jan 2023 – Jun 2023 · Chandigarh

- Led key backend microservices for the launch of **IkarusNest**: authentication, carts, orders and slicer.
- Built email/password and social login with **JWT sessions**, cutting login time by **30%**.
- Added **JWT header-based service-to-service auth** and caching for REST calls, cutting API latency by **50%**.
- Built a **roles and permissions** API, integrated with the UI, that secures all **7+** backend services.
- Deployed 7+ backend services and UI frontends (React, Next.js, FastAPI); added **Liquibase** migrations to the auth MySQL database, cutting manual schema-change time by **20%**; ran three **gRPC** proofs of concept for moving services off REST.

VENTURES & TEACHING

Chukta · Solo builder, side project · getchukta.in

Oct 2026 - Present · Android beta

- A **free, ad-free Splitwise alternative** for friends in India: **one-tap UPI settle-up**, receipt scanning with item-by-item splits, offline use, Splitwise import, WhatsApp reminders and pay-me links. **Android beta live** (Oct 2026); iPhone web app at app.getchukta.in.
- One **Expo / React Native** codebase (TypeScript, NativeWind) for the Android app and the iPhone PWA, on **Supabase**: Postgres with row-level security, Auth, Edge Functions and pg_cron.
- Money in **integer paise**, never floats. Every split is recomputed on the server by a PL/pgSQL port of the TypeScript split logic; a differential test compares the two on **2,000** random groups to the paisa, alongside pgTAP, Vitest and fast-check tests.
- **On-device OCR** (ML Kit, Apple Vision, Tesseract.js on web) with an optional AI check verified to the paisa; a **replay-safe offline outbox**; a Splitwise import that rolls back unless every balance matches the file.

Markdrop · Founder, side venture · markdrop.in

Mar 2026 - Present

- Publish markdown and AI docs with no account, or upload HTML, PDFs, sheets, Word docs, video and zipped sites: **one link that renders in any browser**.
- Web app, **VS Code extension** and a **CLI on Homebrew and Scoop**; TypeScript and Next.js, designed and built solo.

System Design Mastery · free course · himanshukukreja.in/learn

- Free course for backend engineers with **1,000+** quick-check questions, plus long-form **engineering case files**.

OPEN SOURCE & PROJECTS

ios-web-streamer WebRTC · Stream and control a real iOS device from a browser.

obscura TypeScript · HLS · Private video: self-hostable HLS from S3 with encrypted segments.

waygate Go · Self-hosted tunnel for exposing local services through your own infra.

codeflow-analyzer npm · CLI that statically analyzes React and Next.js codebases into behavior reports.

- **LogIngestor** Nov 2023 · Scalable log ingestion on FastAPI, RabbitMQ, MongoDB and Elasticsearch, with JWT auth, asyncio concurrency and rate limiting; Docker on AWS ECS and ECR.
- **AI-powered smart glasses** Oct – Dec 2022 · team of 5 · For visually impaired people: face recognition, image captioning, currency and traffic/object detection on a NodeMCU ESP8266 with Python and TensorFlow.
- **SmartAgri Solutions** Mar 2022 · Crop recommendations with a Random Forest at **96%** accuracy, fertilizer prediction and TensorFlow plant-disease detection (up to 80%); Flask on GCP.

SKILLS

LANGUAGES	Python, TypeScript, JavaScript, Go, SQL, Bash, C++
BACKEND	FastAPI, Flask, Django, Node.js, Bun, microservices, REST, gRPC, WebSockets, JWT and social auth
DATA	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, RabbitMQ, Supabase, Liquibase
CLOUD & INFRA	AWS (ECS, EFS, ECR), GCP Cloud Run, Docker, CI/CD, GitHub Actions, Nginx, Terraform, Ansible, Linux
REALTIME & DEVICES	WebRTC, Android (ADB) and iOS (simctl, idb) device control, secure TCP tunnels, HLS video
AI & AGENTS	LLM agents, MCP, Playwright automation, on-device OCR, TensorFlow
MOBILE & FRONTEND	React Native, Expo, React, Next.js, Electron, Tailwind / NativeWind, PWAs, CLIs and VS Code extensions
TESTING	Vitest, fast-check (property tests), pgTAP, Playwright

EDUCATION

B.E. Computer Science · Thapar Institute of Engineering & Technology

Aug 2019 – Jul 2023 · CGPA **9.04**/10

COURSES

Using Databases with Python
University of Michigan · Sep – Dec 2022

Web Application Technologies and Django
University of Michigan · May – Jul 2020

AWARDS

3rd place, Web Set Go Hackathon, Thapar University.

Inspire Award, State Level Exhibition and Project Competition, Chandigarh University.

10th rank in the Punjab state matriculation exams among 3.3 lakh students.